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Mark Dynrad Luistro Maranan

Nationality: Italian, Philippine **Date of birth:** 06/10/2003 **Gender:** Male

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ABOUT ME

Bachelor's Degree in Computer Engineering in October 2026 (Ingegneria Informatica) at Università La Sapienza di Roma.

Strongly experienced in Applied Computer Science, Embedded Systems and Electronics.

Positive attitude, well-disciplined, strong-minded, willing and motivated to show my skills and acquire new ones. I love getting involved in problem solving scenarios and I'm constantly searching for new challenges. Fast learner.

Full of energy, capable of multitasking, proficient in working in groups and under pressure.

WORK EXPERIENCE

FAST CHARGE ENGINEERING SRL – Roma, Italy

EMBEDDED SYSTEM SOFTWARE DEVELOPER

[16/10/2024 – 10/12/2024]

- Development of software for high-performance Battery Management Systems (BMS)
- Assistance in PCB review and verification
- Execution of SIL (Software-in-the-Loop) and HIL (Hardware-in-the-Loop) testing

- **Events:**
  - Zero Emission, Rome, Italy (2024)

SAPIENZA FAST CHARGE UNIVERSITY RACING TEAM – Roma, Italy

SAPIENZA FAST CHARGE TEAM MEMBER

[11/2022 – 2026]

- TR (Technical Responsible, 2023 - 2026):
  - Analysis, design and review of software for data management, PCB, sensing systems.
  - Analysis and application of both wired and wireless communication protocols
  - Work planning and resources management
  - Addressing new recruits

- Projects:
  - **Telemetry System:** Design of the pit-to-car (Car-To-Box) communication architecture and development of the driver-vehicle interface
  - **Battery Management System (BMS):** Analysis and firmware development for High-Voltage BMS, along with design, assembly, and review of custom PCBs
  - **Electronics:** Requirements analysis, design, and assembly of onboard electronic PCBs

- **Firmware:** Analysis and programming for the Vehicle Control Unit (VCU)
- **Driverless Department:** Technical support and hardware/software development

- Events:
  - Formula FATA Italy 2023
  - Formula FSS Spain 2026
  - Maker Faire Italy 2025

EDUCATION AND TRAINING

COMPUTER ENGINEERING BACHELOR DEGREE

Università La Sapienza [09/2022 – 2026]

City: Roma | **Country:** Italy | **Website:** https://www.uniroma1.it/ | **Final grade:** 104 | **Thesis:** Progettazione di un sistema hardware/software di diagnostica, acquisizione dati e telemetria real-time di una vettura da corsa della Fomula SAE

- **BMS Software Development:** Design and implementation of software for high-voltage battery management
- **Telemetry and Custom Board Development:** Firmware programming for telemetry systems and design of custom PCB boards
- **HMI Optimization:** Development and enhancement of the driver-machine interface
- **Aero Rake HW/SW Integration:** Hardware and software integration for aerodynamic sensing systems

QUALIFICATION AS ELECTRICAL OPERATOR (PES-PAV-PEI)

CENTRO STUDI ATHENA SRL [2024]

HIGH SCHOOL DIPLOMA

IIS VIA SILVESTRI 301 (Liceo Scientifico Marcello Malpighi) [09/2017 – 06/2022]

City: Roma | **Country:** Italy | **Website:** https://www.iisviasilvestri301roma.edu.it/ | **Final grade:** 99 / 100

CAMBRIDGE FIRST CERTIFICATE IN ENGLISH

CAMBRIDGE ASSESSMENT English [2020]

City: Roma | **Country:** Italy

CAMBRIDGE ASSESMENT PHYSICS - CAMBRIDGE ASSESMENT GEOGRAPHY/ENGLISH AS A SECOND LANGUAGE (IGCSE)

IIS VIA SILVESTRI 301 (Liceo Scientifico Marcello Malpighi) [2019]

City: Roma | **Country:** Italy

ECDL CERTIFICATE

[2017]

LANGUAGE SKILLS

Mother tongue(s): Italian | Tagalog

Other language(s):

English

LISTENING C1 **READING** C1 **WRITING** C1

SPOKEN PRODUCTION C1 **SPOKEN INTERACTION** C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

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SKILLS

Programming languages and Frameworks/OS

C / C++ / C/C++ for Microcontrollers / Python (computer programming) / MicroPython / CircuitPython / Assembly (computer programming) / Java (computer programming) / MATLAB / MATLAB, MATLAB Simulink / Arduino / ESP-IDF / STMCube IDE / SPC5 Studio / FreeRTOS / Operation Systems Windows, Linux / ROS, ROS2

Software

ALTIUM DESIGNER / VS/VS Code / Nextion Editor / PCAN View / BusmasterCan Analyzer / Electronic Hardware Design: EasyEDA / Foxglove Studio / use microsoft office

SOFT SKILLS

TIME AND STRESS MANAGEMENT

FLEXIBILITY

CREATIVITY

INTERPERSONAL AND COMUNICATIVE SKILLS

CRITICAL THINKING AND SOLVING

TEAM WORKING

WORK PLANNING

FUNDRAISING AND NETWORKING

HARD SKILLS

EMBEDDED PROGRAMMING

- **Strong proficiency in microcontrollers and integrated circuits** (logic gates, digital ICs, and analog sensors)
- **Extensive experience with low-level peripherals and communication protocols** (CAN bus, SPI, isoSPI, UART, I2C, I2S)
- **Development of complex hardware and firmware projects:**
    - Telemetry system for a Formula SAE team based on the ESP32-S3 microcontroller (dual-core Xtensa)
    - BMS software for the SPC58EC microcontroller (32-bit ST Power Architecture) and custom dedicated boards
    - Custom VGA graphics driver built entirely on ESP32
- **Implementation of P2P communication devices** using Morse code and the MQTT protocol

PCB DEVELOPMENT

- **Proficiency in Simulation Software:** LTspice, Falstad, and Logisim
- **Component Selection & Datasheet Analysis:** In-depth knowledge of major distributors/suppliers and strong ability to navigate and analyze complex datasheets
- **Advanced PCB Design:** Expert of EDA tools for electronic design, such as Altium Designer and EasyEDA
- **Complex PCB Development:** Extensive experience in multi-layer PCB design (single, double, and up to 4 layers)

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FULL STACK DEVELOPER

- **Full Stack Development:** Intermediate level
- **Back-End Development:** Intermediate proficiency in FastAPI, wireless communication protocols, and cybersecurity
- **Front-End Development:** Basic knowledge of React and JavaScript

MANUAL PCB SOLDERING

- **Experience in soldering THT and SMD components** on PCBs of various sizes, with and without stencils
- **Techniques and equipment used:**
    - Hand soldering (soldering iron)
    - Hot air rework/soldering
    - Hot plate soldering

HOMELABBING

- **Custom NAS Deployment:** Built a personal NAS solution based on an Intel Single Board Computer
- **Home Networking & Smart Home Integration:** Configuration of local networks and integration of IoT/smart devices
- **Web App Hosting:** Deployment and management of self-hosted web applications for testing and skill development
- **Local Machine Learning:** Experimentation and execution of foundational AI/ML models on local hardware for learning and research purposes
- **Hardware Modding:** Strong passion for modding, customizing, and repairing handheld devices and gaming consoles